

CLINICAL FOUNDATION

Intake, Pattern Recognition & Testing Roadmap

THE BRIDGE: FROM THEORY TO PRACTICE

You now understand the systems-biology framework - the axes, the feedback loops, the interconnections. But how does that translate into a real clinical encounter?

This module is the bridge. Before we go deep into each hormone system, we need a shared clinical language: how to **listen** for patterns, how to **differentiate** between axis, gland, and receptor problems, and how to build a **testing roadmap** that matches the clinical picture.

The intake is not a form. It is a diagnostic instrument.

1

Listen: Recognize symptom clusters and axis patterns



2

Differentiate: Axis vs. Gland vs. Receptor dysfunction



3

Map: Build a targeted testing roadmap



4

Interpret: Functional ranges, not just "normal"

THE HORMONE INTAKE PHILOSOPHY

Most practitioners collect **symptoms**.

Functional hormone practitioners listen for **patterns**.

A symptom is a data point. A pattern is a story — and that story points to a specific axis, a specific mechanism, and a specific testing strategy.

STANDARD INTAKE	FUNCTIONAL HORMONE INTAKE
Collects a list of symptoms	Groups symptoms into axis clusters
Asks "What's wrong?"	Asks "Which system is most loaded?"
Treats each symptom separately	Identifies the upstream driver
Orders standard labs	Builds a targeted testing roadmap
Interprets "normal" ranges	Interprets functional/optimal ranges

The goal of the intake is not to document complaints — it is to identify *which system is under the most load* and *where in that system* the breakdown is occurring.

THE CLINICAL INTAKE: QUESTIONS THAT DRIVE DECISIONS

Ask open-ended questions that reveal patterns — not just symptoms

HPA AXIS — STRESS & ADRENALS

- " Walk me through a typical day — when do you feel best, and when do you feel worst? *(Maps the cortisol curve without ordering a single test)*
- " What happens when you try to rest or take a day off — do you feel restored, or do you feel guilty and restless? *(Reveals nervous system dysregulation and cortisol patterning)*
- " How long have you been running at this pace? Has there been a major stressor — physical, emotional, or life event — in the last 1–3 years? *(Identifies allostatic load timeline)*

HPG AXIS — CYCLING HORMONES

- " Tell me about your cycle — how long is it, how heavy is the flow, and how do you feel in the week before your period? *(The cycle is a monthly hormonal report card)*
- " Would you say you feel like a different person in the week before your period? *(Screens for PMDD/luteal phase progesterone deficiency)*
- " How has your libido changed over the last few years — and does it track with your cycle at all? *(Low libido + no mid-cycle peak = anovulatory or low testosterone)*

HPT AXIS — THYROID

- " Are you always the cold one in the room? Do you wear socks to bed or need a blanket when others are comfortable? *(Cold intolerance is one of the most reliable hypothyroid indicators)*
- " Has your digestion slowed down — are you going to the bathroom less than once a day? *(Gut motility is thyroid-dependent; constipation is a key functional marker)*
- " Have you ever been told your thyroid is "normal" but you still feel like your metabolism has stopped? *(Opens the door to functional vs. standard range conversation)*

METABOLIC CORE — INSULIN & BLOOD SUGAR

- " What happens if you skip a meal or delay breakfast by a couple of hours — how do you feel? *(Reveals metabolic flexibility and glucose dependency)*
- " Do you notice a significant energy drop or brain fog after eating — especially after a carbohydrate-heavy meal? *(Post-prandial glucose dysregulation pattern)*
- " Where does your body tend to store weight — and has that changed? Has your waistline grown even when your overall weight hasn't? *(Visceral fat = cortisol + insulin interaction)*

Practitioner Skill: The best intake questions are open-ended, non-leading, and invite the client to tell their story — then you listen for the axis pattern underneath.

MOTIVATIONAL INTERVIEWING & READINESS TO CHANGE

Your clinical knowledge is only as useful as your client's readiness to act on it

THE OARS FRAMEWORK

O Open Questions Invite exploration, not yes/no answers. "What matters most to you about your health right now?"	A Affirmations Acknowledge strengths and efforts genuinely. "You've clearly been paying close attention to your body."
R Reflections Mirror back what you hear — especially ambivalence. "It sounds like you want to feel better, but you're not sure where to start."	S Summaries Collect and reflect key themes at the end of a section. Builds trust and shows you were listening.

ASSESSING READINESS: TWO QUESTIONS

IMPORTANCE	"On a scale of 1–10, how important is it to you to address this right now?" — If below 7, explore the ambivalence before building a protocol.
CONFIDENCE	"On a scale of 1–10, how confident are you that you could make these changes?" — Low confidence = the plan is too big. Break it down.

LANGUAGE THAT BUILDS VS. BREAKS TRUST

AVOID (EXPERT TRAP)	USE INSTEAD (MI SPIRIT)
"You need to stop eating sugar."	"What would it look like for you to reduce sugar in a way that felt manageable?"
"Your cortisol is high because of stress."	"What do you think has been the biggest drain on your energy over the last year?"
"You should be in bed by 10 PM."	"What gets in the way of winding down earlier — and what would need to change for that to feel possible?"
"Here's your protocol — follow this."	"Of everything we've discussed, what feels like the most important place to start for you?"
"Why haven't you done this yet?"	"What's made it hard to prioritize this until now?"

The Expert Trap: Telling clients what to do activates resistance. Asking them to articulate their own reasons for change activates motivation. Your job is to evoke — not prescribe.

SETTING REALISTIC EXPECTATIONS: TIMELINES & HEALING ORDER

Clients who understand the process stay in the process

THE HEALING ORDER — WHY SEQUENCE MATTERS

YOU CANNOT SKIP STEPS



Chronic stress suppresses thyroid conversion and sex hormone production. Insulin resistance worsens estrogen dominance. Downstream systems need upstream drivers addressed first.

APPROXIMATE TIMELINES

Weeks 1–4

Foundation: sleep, stress load, blood sugar stability. Client notices improved energy and sleep quality.

Months 3–6

Thyroid conversion improving. Cycle symptoms reducing. Body composition beginning to shift.

Months 1–3

Metabolic stabilization, HPA rhythm improving. Labs begin to shift. Mood and cognition improve.

Months 6–12+

Full HPG axis rebalancing. Sustainable hormone optimization. BHRT consideration if indicated.

WHAT TO SAY TO CLIENTS

💬 **"This is a process, not a prescription."** We are rebuilding the foundation of your health — that takes months, not days. The goal is sustainable change, not a quick fix.

💬 **"You may feel worse before you feel better."** As the body starts to shift, symptoms can temporarily intensify — especially fatigue and mood. This is a sign of change, not failure.

💬 **"We address the root cause first."** If stress is driving your thyroid and hormone issues, no amount of thyroid or hormone support will hold until the stress response is stabilized.

💬 **"Progress is not always linear."** There will be good weeks and harder weeks. We track trends over months — not day-to-day fluctuations.

Clinical Rule: A client who understands *why* the process takes time is far more likely to stay committed than one who expects results in 2 weeks.

THE INTAKE FORMS: HOW TO USE THEM IN PRACTICE

The HPATG Questionnaire translates symptoms into a testing and treatment roadmap

WHAT THE FORM COVERS — SECTION BY SECTION

HPA AXIS	Stress, Energy & Sleep Screens for cortisol curve patterns, HPA load, and sleep architecture disruption. High scores here = start with cortisol testing (DUTCH or saliva) before anything else.
METABOLIC CORE	Insulin, Blood Sugar & Body Composition Identifies glucose dysregulation, insulin resistance patterns, and metabolic flexibility. High scores = fasting insulin, HbA1c, and CGM trial are priority.
HPT AXIS	Thyroid Function Screens for hypothyroid symptom burden — cold intolerance, hair loss, constipation, brain fog. High scores = full thyroid panel including Free T3, Reverse T3, and antibodies.
HPG AXIS	Sex Hormones & Cycle Health Maps estrogen, progesterone, and androgen symptom patterns. Includes cycle tracking data. High scores = DUTCH (luteal phase) + serum sex hormone panel.
LIFESTYLE LOAD	Sleep, Nutrition, Movement & Environment Identifies the behavioral and environmental drivers of dysfunction. Informs the lifestyle protocol alongside testing.

HOW TO USE IT: 4-STEP WORKFLOW

- 1 Send before the first session.** Review responses before you meet — come in already knowing which axis is most loaded and what questions to ask.
- 2 Identify the dominant axis.** Which section has the most "often" or "always" responses? That axis drives your initial testing and intervention priority.
- 3 Note cross-axis patterns.** Symptoms appearing across multiple axes signal systemic dysregulation — likely HPA-driven. Don't treat each axis in isolation.
- 4 Use it as a conversation opener.** "I noticed you rated your sleep a 3 out of 5 and your energy a 2 — tell me more about what a typical day looks like for you."

When all four axes are loaded: This is a systemic HPA-driven picture. Start with stress, sleep, and blood sugar stabilization before ordering a full hormone panel. Testing an exhausted, dysregulated system gives you a snapshot of dysfunction — not a baseline.

Remember: The form is a starting point — not a diagnosis. Use it to direct your questions, not replace them.

LISTENING FOR HPA AXIS PATTERNS

HPA / ADRENAL

These symptom clusters point toward HPA axis dysfunction — a dysregulated stress response, not simply "adrenal fatigue."

TIRED BUT WIRED

Exhausted during the day but can't wind down at night. Racing thoughts at bedtime. Feels "second wind" after 9 PM.

Suggests: Elevated evening cortisol — disrupted diurnal rhythm, often Stage 1–2 HPA dysregulation

MORNING FATIGUE

Unrefreshed on waking. Needs 1–2 hours to "come online." Worst energy is in the morning.

Suggests: Blunted or absent Cortisol Awakening Response (CAR) — low morning cortisol output

SALT CRAVINGS

Craves salty foods. Adds salt to everything. Feels better after salty meals. Low blood pressure.

Suggests: Low aldosterone / reduced mineralocorticoid activity — sodium wasting pattern

2–4 AM WAKING

Wakes between 2–4 AM with racing mind or anxiety. Difficulty returning to sleep. Often hungry.

Suggests: Nocturnal cortisol spike or reactive hypoglycemia — blood sugar dysregulation driving cortisol release

AFTERNOON CRASH

Significant energy drop between 2–4 PM. Needs caffeine or sugar to push through. Brain fog after lunch.

Suggests: Blunted mid-afternoon cortisol — flat diurnal curve, often with insulin dysregulation

ORTHOSTATIC SYMPTOMS

Dizzy or lightheaded on standing. Postural hypotension. Feels worse in heat. Palpitations on exertion.

Suggests: Reduced cortisol/aldosterone support for vascular tone — consider POTS overlap

Key Question to Ask: "When during the day do you feel worst — and when do you feel best?" The timing tells you the cortisol curve.

LISTENING FOR HPT AXIS PATTERNS

HPT / THYROID

These symptom clusters suggest thyroid axis dysfunction — often subclinical hypothyroidism, conversion issues, or autoimmune thyroiditis.

COLD INTOLERANCE

Always cold, especially hands and feet. Wears socks to bed. Feels cold when others are comfortable. Low basal body temperature.

Suggests: Reduced metabolic rate — low Free T3 or poor T4 → T3 conversion

OUTER EYEBROW LOSS

Thinning or loss of the outer third of the eyebrow (Queen Anne's sign). Diffuse hair thinning or shedding. Dry, coarse hair.

Suggests: Hypothyroid pattern — classic physical exam finding; also check TPO antibodies

SLUGGISH BOWELS

Less than one bowel movement per day. Constipation, hard stools. Bloating. Slow gastric emptying. Feels "backed up."

Suggests: Reduced gut motility from low thyroid hormone — T3 drives peristalsis

WEIGHT RESISTANCE

Difficulty losing weight despite caloric restriction and exercise. Weight gain without dietary changes. Puffy, not just fat.

Suggests: Reduced basal metabolic rate — also consider Reverse T3 dominance under chronic stress

BRAIN FOG / MEMORY

Difficulty with word recall, concentration, or short-term memory. Feels mentally "slow." Difficulty learning new things.

Suggests: Reduced T3 availability to the brain — T3 is essential for neuronal function and myelination

MORNING STIFFNESS

Joint and muscle stiffness on waking that improves through the day. Aching without injury. Carpal tunnel symptoms.

Suggests: Hypothyroid myopathy — mucopolysaccharide accumulation in tissues; also check TPO/TgAb

Key Question to Ask: "Are you always the cold one in the room? Do you feel like your metabolism has just stopped?" — The body temperature story is diagnostic.

LISTENING FOR HPG AXIS PATTERNS

HPG / SEX HORMONES

These symptom clusters suggest HPG axis dysfunction – estrogen/progesterone imbalance, androgen excess, or declining ovarian function.

PMS / PMDD

Mood swings, irritability, crying spells, anxiety in the 7–10 days before menstruation. Symptoms resolve with period onset.

Suggests: *Low luteal progesterone or estrogen dominance – also consider impaired estrogen metabolism (4-OH pathway)*

HEAVY / PAINFUL PERIODS

Soaking pads/tampons hourly, passing clots, severe cramping. Periods lasting more than 7 days. Iron deficiency anemia.

Suggests: *Estrogen dominance with relative progesterone deficiency – also rule out fibroids, endometriosis*

ANDROGENIC SYMPTOMS

Jawline/chin acne, unwanted facial hair (hirsutism), scalp hair thinning at the crown, oily skin.

Suggests: *Androgen excess – elevated testosterone, DHEA-S, or 5 α -reductase activity; consider PCOS*

VASOMOTOR SYMPTOMS

Hot flashes, night sweats, sudden waves of heat. Worse at night. Disrupts sleep. Can occur in perimenopause or with low estrogen at any age.

Suggests: *Declining estradiol – loss of thermoregulatory stability in the hypothalamus*

LOW LIBIDO

Significantly reduced sexual desire. Difficulty with arousal. Reduced genital sensitivity. Often accompanied by fatigue and mood changes.

Suggests: *Low testosterone and/or low estradiol – also consider HPA axis load suppressing HPG axis*

IRREGULAR CYCLES

Cycles shorter than 24 or longer than 35 days. Skipped periods. Spotting between periods. Unpredictable cycle length.

Suggests: *Anovulatory cycles, perimenopause, PCOS, or HPA suppression of GnRH pulsatility*

Key Question to Ask: "Tell me about your cycle – the timing, the flow, and how you feel in the week before your period." The cycle is a monthly report card.

LISTENING FOR METABOLIC CORE PATTERNS

INSULIN / METABOLIC

"HANGRY"

Irritable, shaky, or lightheaded if meals are delayed. Mood dramatically improves after eating. Can't skip meals.

Suggests: Reactive hypoglycemia — poor glucose buffering, early insulin resistance

POST-MEAL CRASHES

Sleepy or foggy 1–2 hours after eating, especially carbohydrate-heavy meals. Needs to lie down after lunch.

Suggests: Post-prandial glucose spike and crash — insulin dysregulation, impaired glucose disposal

BELLY FAT / WAIST

Weight concentrated around the midsection. Waist circumference > 35 inches. "Apple shape" body composition change.

Suggests: Visceral adiposity — driven by elevated cortisol + insulin; high cardiovascular risk marker

SKIN TAGS / ACANTHOSIS

Small skin tags (neck, armpits, groin). Dark, velvety patches of skin (acanthosis nigricans) at neck or skin folds.

Suggests: Insulin receptor overstimulation — a physical sign of significant insulin resistance

CARB CRAVINGS

Strong cravings for sweets, bread, pasta. Feels compelled to eat after dinner. Difficult to feel satisfied with protein/fat meals.

Suggests: Glucose dependency — impaired fat oxidation, mitochondrial inflexibility

OVERNIGHT FASTING DIFFICULTY

Cannot go 12+ hours without eating. Wakes hungry at night. Feels unwell if breakfast is delayed.

Suggests: Poor metabolic flexibility — unable to shift to fat burning when glucose is unavailable

WHY METABOLIC CORE MATTERS

Insulin resistance is the **silent amplifier** of every other hormonal dysfunction. It worsens HPA dysregulation, suppresses thyroid conversion, drives androgen excess in PCOS, and accelerates estrogen dominance.

The metabolic core must always be assessed — even when the chief complaint is "hormonal." You cannot fully resolve a hormone imbalance in an insulin-resistant patient.

Key Question: "What happens if you skip a meal? Can you go 12 hours without eating?" The answer reveals metabolic flexibility.

AXIS VS. GLAND VS. RECEPTOR

Three different problems — three different clinical strategies

LEVEL 1 — TOP-DOWN AXIS / BRAIN PROBLEM

WHAT IT MEANS

The hypothalamus or pituitary is not sending the right signal. The gland itself may be healthy, but it is not being told to produce.

Example (Thyroid): *TSH is low-normal, Free T4 is low. The brain is not stimulating the thyroid. Often seen in chronic stress, HPA dysregulation, or caloric restriction.*

Look for: Low or "inappropriately normal" stimulating hormone (TSH, LH, FSH) with low end-organ output

LEVEL 2 — GLAND FAILURE PRIMARY GLAND PROBLEM

WHAT IT MEANS

The brain is signaling correctly, but the gland itself is damaged, depleted, or autoimmune-attacked and cannot respond adequately.

Example (Thyroid): *TSH is elevated (brain working hard), but Free T4 and Free T3 are low. The thyroid gland is failing. Classic Hashimoto's or primary hypothyroidism.*

Look for: Elevated stimulating hormone (TSH, LH, FSH) with low end-organ output — and consider antibodies

LEVEL 3 — CELLULAR RECEPTOR RESISTANCE

WHAT IT MEANS

The brain is signaling, the gland is producing, but the cells cannot use the hormone. Hormone levels may appear "normal" on labs.

Example (Thyroid): *TSH, Free T4, Free T3 all appear normal — but patient has all hypothyroid symptoms. Consider cellular T3 resistance, inflammation, or nutrient cofactor deficiency.*

Look for: "Normal" labs with persistent symptoms — test Reverse T3, inflammatory markers, nutrient status

Clinical Rule: Always ask — is this a signaling problem, a production problem, or a utilization problem? The answer determines the intervention.

THE TESTING ROADMAP: AN OVERVIEW

1

TEST TO CONFIRM, NOT TO DIAGNOSE

The intake and pattern recognition come first. Testing confirms the clinical hypothesis — it does not replace clinical thinking.

2

TIMING MATTERS

Hormones are not static. Cortisol, sex hormones, and thyroid hormones all have optimal collection windows. Testing at the wrong time produces misleading data.

3

NORMAL ≠ OPTIMAL

Standard lab ranges are built on population averages that include sick people. Functional reference ranges represent the levels associated with *optimal health outcomes* — not simply the absence of disease.

4

MATCH THE TOOL TO THE QUESTION

Each testing modality answers a different clinical question. Serum, DUTCH, saliva, and wearables are not interchangeable — they are complementary.

THE FUNCTIONAL TESTING TOOLKIT



Serum / Blood Testing

Baseline hormones, thyroid panel, metabolic markers, antibodies



DUTCH (Dried Urine)

Cortisol curves, estrogen metabolites, androgen pathways



Salivary Testing

Free cortisol rhythm, Cortisol Awakening Response (CAR)



Wearables & CGM

HRV, sleep architecture, continuous glucose monitoring

SERUM TESTING: THE BASELINE

TOOL 1 OF 4

WHAT TO ORDER — BY AXIS

HPA

Adrenal / Stress Panel

DHEA-S, morning cortisol (8 AM), pregnenolone

HPT

Full Thyroid Panel

TSH, Free T4, Free T3, Reverse T3, TPO antibodies, TgAb

HPG

Sex Hormone Panel

Estradiol, progesterone (Day 21), total & free testosterone, FSH, LH, SHBG, prolactin

METABOLI

Metabolic Core Panel

Fasting insulin, fasting glucose, HbA1c, lipid panel (with TG:HDL ratio), hs-CRP, ferritin, vitamin D

STRENGTHS & LIMITATIONS

- ✓ Best for: baseline hormone levels, antibodies, metabolic markers, thyroid panel — the essential first-pass assessment
- ✗ Does not show hormone metabolites (how estrogen is being broken down)
- ✗ Single snapshot — misses diurnal variation in cortisol (one morning cortisol is not a cortisol curve)
- ✗ SHBG can obscure free hormone availability — always order free fractions alongside total

Timing Rules:

HPA: Morning cortisol at 8 AM, fasting

HPT: Any time, fasting preferred; do not take thyroid meds before draw

HPG: Day 3 for FSH/LH/E2 baseline; Day 21 for progesterone peak

Metabolic: Fasting 10–12 hours for insulin and glucose

DUTCH TESTING: THE METABOLITE MAP

TOOL 2 OF 4

WHAT DUTCH MEASURES — BY CATEGORY

CORTISOL RHYTHM

HPA Axis & Diurnal Cortisol Curve

5-point cortisol collection (waking, 30 min post-waking, noon, evening, bedtime) + Cortisol Awakening Response (CAR). Total metabolized cortisol (THF + THE) to assess production vs. clearance.

ESTROGEN METABOLISM

Estrogen Metabolite Pathways

2-OH estrone (protective), 4-OH estrone (genotoxic), 16-OH estrone (proliferative). 2-methoxy estrone (COMT activity). Essential for breast cancer risk assessment and estrogen dominance.

ANDROGEN PATHWAYS

Androgen Production & Conversion

DHEA, testosterone, DHT (5 α -reductase activity), androsterone. Identifies whether androgen excess is from production or conversion.

PROGESTERONE

Progesterone Metabolites

Pregnanediol and allopregnanolone (ALLO) — the calming, GABA-A agonist metabolite. Low ALLO is associated with anxiety, poor sleep, and PMDD.

STRENGTHS & LIMITATIONS

- ✓ Only test that shows the full cortisol curve AND estrogen metabolites in one collection
- ✓ Reveals *how* hormones are being metabolized — not just how much is present
- ✓ Identifies COMT and 5 α -reductase enzyme activity patterns
- ✗ Does not replace serum for TSH, thyroid antibodies, or metabolic markers
- ✗ Affected by exogenous hormone use — must note all supplements and medications

Timing Rules:

Cycling: Days 19–22 of cycle (luteal phase) for sex hormones

Cortisol: Any time — but note sleep/wake times accurately

Menopause: Any time (no cycling pattern)

WHY SALIVA?

Saliva measures the **free, bioavailable fraction** of cortisol — the portion that has actually crossed into tissues and is biologically active. Serum cortisol is mostly protein-bound (to CBG) and does not reflect what cells are actually experiencing.

BEST USED FOR:

- ✓ **Cortisol Awakening Response (CAR):** The 50–160% rise in cortisol from waking to 30 minutes post-waking. A blunted CAR is one of the most consistent findings in HPA dysregulation.
- ✓ **Diurnal cortisol rhythm:** 4-point collection (waking, noon, evening, bedtime) to map the daily cortisol curve pattern.
- ✓ **Free sex hormones:** Salivary estradiol, progesterone, testosterone, and DHEA reflect the free, unbound fraction — useful when SHBG is high and serum totals are misleading.
- ✗ Does not measure cortisol metabolites or hormone pathways (use DUTCH for total production and estrogen metabolism)
- ✗ Does not measure thyroid markers — serum remains the standard for TSH, Free T4, Free T3, and antibodies

SALIVA VS. DUTCH FOR CORTISOL

FEATURE	SALIVA	DUTCH
Measures	Free cortisol + free sex hormones	Free + metabolized cortisol + hormone pathways
CAR Accuracy	Gold standard	5-point DUTCH also captures CAR
Total Production	No	Yes (THF + THE)
Cost	Lower	Higher
Best For	Free hormone fractions; cortisol rhythm; SHBG-adjusted sex hormones	Hormone metabolites, estrogen pathways, androgen conversion

The CAR is a window into HPA resilience. A healthy CAR signals the brain is mounting an appropriate morning stress response. A flat CAR suggests the HPA axis is no longer able to respond adequately to daily demands.

WEARABLES & CONTINUOUS MONITORING

TOOL 4 OF 4

Wearables provide continuous, real-world data that no single lab draw can capture — bridging the gap between the clinic and daily life.

CONTINUOUS GLUCOSE MONITOR CGM

WHAT IT MEASURES

Interstitial glucose every 5 minutes, 24 hours a day. Captures post-meal spikes, overnight patterns, and dawn phenomenon.

CLINICAL VALUE

Reveals metabolic flexibility — how well the patient handles carbohydrates, whether they experience reactive hypoglycemia, and how their glucose responds to stress and sleep.

Key insight: A patient with "normal" fasting glucose may have significant post-meal spikes that are driving cortisol dysregulation and fat storage.

OURA RING / WHOOP / GARMIN HRV & SLEEP TRACKING

WHAT IT MEASURES

Heart Rate Variability (HRV), resting heart rate, sleep stages (deep/REM/light), respiratory rate, and recovery scores.

CLINICAL VALUE

HRV is a proxy for HPA axis resilience — low HRV reflects high allostatic load. Declining HRV trends often precede clinical symptoms of burnout and HPA dysregulation.

Key insight: A patient with subjectively "good" sleep but consistently low HRV and poor deep sleep is not recovering — their nervous system is still in a stress response.

BASAL BODY TEMPERATURE BBT TRACKING

WHAT IT MEASURES

Resting temperature taken immediately on waking (before getting up). Rises 0.2–0.5°C after ovulation due to progesterone's thermogenic effect.

CLINICAL VALUE

Confirms ovulation without a blood draw. A consistently low BBT (<97.4°F / 36.3°C) suggests hypothyroid pattern or anovulation. No mid-cycle rise = no progesterone production.

Key insight: BBT charting over 2–3 cycles provides more clinical information than a single Day 21 progesterone draw — it shows the pattern, not just a snapshot.

The Principle: Wearables do not replace lab testing — they provide the *context* that makes lab results meaningful. Always interpret labs alongside lifestyle data.

KEY TAKEAWAYS

YOU NOW HAVE THE CLINICAL FRAMEWORK

- 1** The intake is a diagnostic instrument — listen for axis patterns, not just individual symptoms.
- 2** Every axis has a symptom signature — HPA, HPT, HPG, and Metabolic Core each present with recognizable clusters.
- 3** Always differentiate Axis vs. Gland vs. Receptor — the level of dysfunction determines the intervention strategy.
- 4** Match the testing tool to the clinical question — serum, DUTCH, saliva, and wearables are complementary, not interchangeable.
- 5** Normal is not optimal — always interpret results against functional reference ranges, not population averages.

COMING UP NEXT

MODULE 2: THE HPA AXIS

We go deep into the stress response system — the physiology of cortisol, the stages of HPA dysregulation, and the clinical protocols for restoring adrenal resilience.

MODULE RESOURCES

HPATG Intake Questionnaire — Patient-facing symptom assessment tool

At-A-Glance Testing Guide — What to order, when, and functional reference ranges